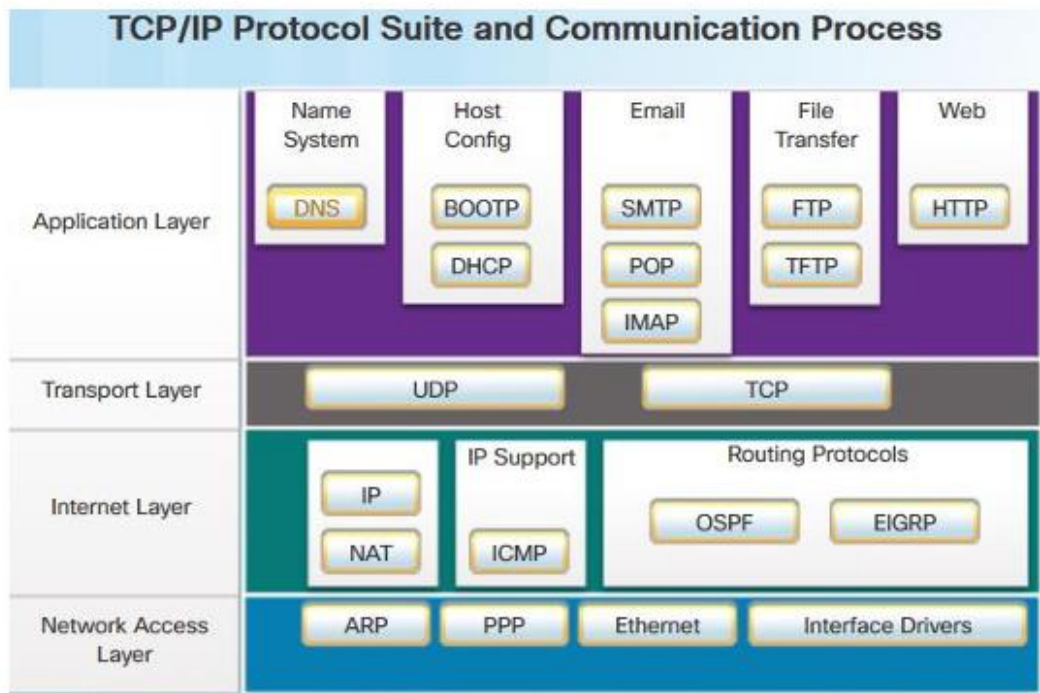


Goal: Understand networking

Need to understand all of the components involved. [1. From the cables that connect devices to each other 2. The protocols that these devices use to communicate.]

Networking protocols define a set of rules for exchanging messages between devices.

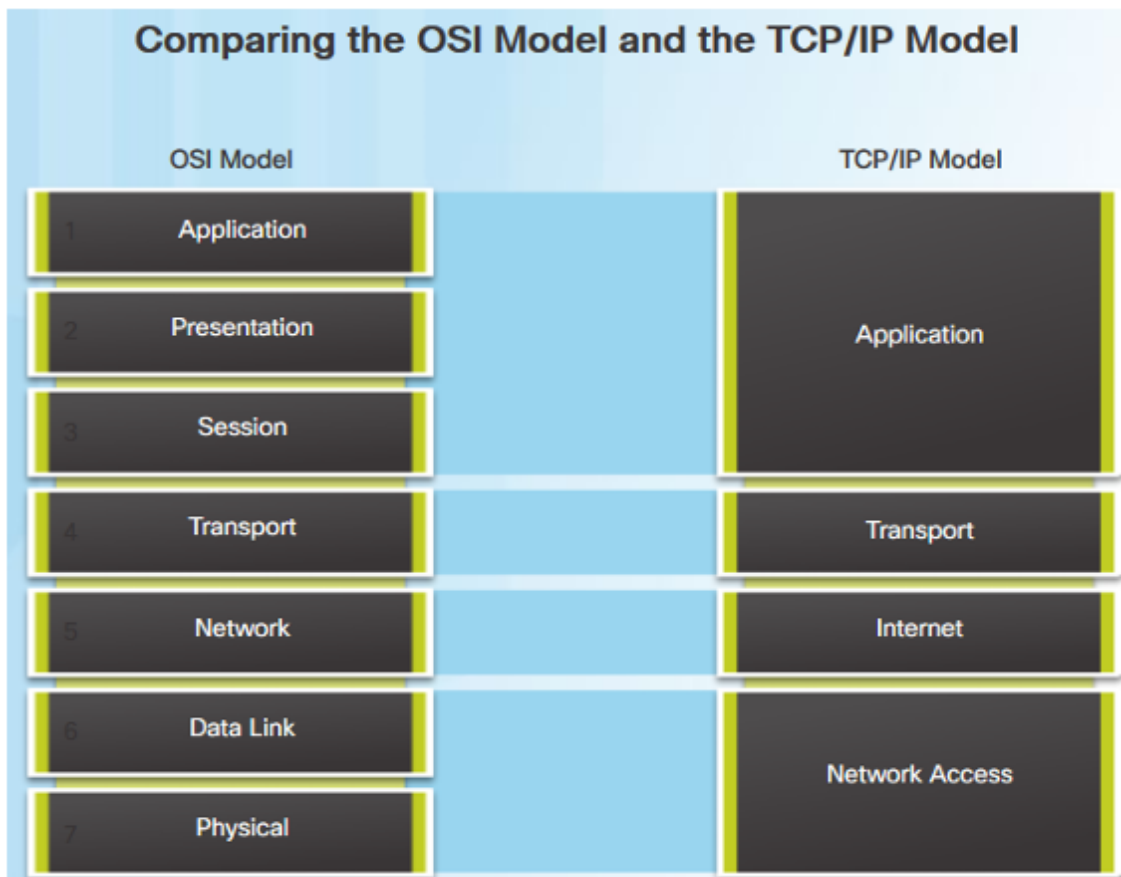
A protocol suite is a set of protocols that work together to provide comprehensive network communication services.



There are many models that help explain how network devices communicate (OSI and TCP/IP model)

Learning outcome

The students should be able to identify and describe each layer and what purpose it serves



#	Layer Name	Protocol	Protocol Data Unit	Addressing
5	Application	HTTP, SMTP, etc...	Messages	n/a
4	Transport	TCP/UDP	Segment	Port #'s
3	Network	IP	Packet / Datagram	IP Address
2	Data Link	Ethernet, Wi-Fi	Frames	MAC Address
1	Physical	n/a	Bits	n/a

Physical Layer

It represents the physical devices that interconnect computers.

This includes the specifications for the networking cables and the connectors that join devices together, along with specifications describing how signals are sent over these connections (cabling, connectors and sending signals).

Data Link Layer

Some sources will call this layer the network interface or the network access layer. At this layer, we introduce our first protocols. The data link layer is responsible for defining a common way of interpreting these signals so network devices can communicate. Lots of protocols exist at the data link layer, but the most common is known as Ethernet, although wireless technologies are becoming more and more popular.

Network Layer

The network layer is also sometimes called the Internet layer.

It's this layer that allows different networks to communicate with each other through devices known as routers. Think of when a device on your home network

connects with a server on the Internet. It's the network layer that helps get the data between these two locations.

The most common protocol used at this layer is known as IP or Internet protocol. IP is the heart of the Internet and most small networks around the world.

The transport layer.

The transport layer sorts out which client and server programs are supposed to get that data.

The protocol most commonly used in the transport layer, is known as TCP or Transmission Control Protocol. Other transport protocols include UDP or user datagram protocol. The big difference between the two is that TCP provides mechanisms to ensure that data is reliably delivered, while UDP does not.

The application layer

The fifth layer is known as the application layer. There are lots of different protocols at this layer, and as you might have guessed from the name, they're application specific.

You can think of layers like different aspects of a package being delivered. The physical layer is the delivery truck and the roads. The data link layer is how the delivery trucks get from one intersection to the next over and over. The network layer identifies which roads need to be taken to to get from address A to address B. The transport layer ensures that delivery driver knows how to knock on your door to tell you your package has arrived, and the application layer is the contents of the package itself.



Physical



Data Link



Network



Transport



Application

The 7 Layers of the OSI Model

- **Layer 7 - Application:** Provides network services directly to user applications (e.g., HTTP, FTP).
- **Layer 6 - Presentation:** Handles data formatting, encryption, and decryption.
- **Layer 5 - Session:** Manages sessions (connections) between applications.
- **Layer 4 - Transport:** Handles end-to-end communication, flow control, and error correction (e.g., TCP, UDP).
- **Layer 3 - Network:** Manages routing and logical addressing (e.g., IP addresses).
- **Layer 2 - Data Link:** Manages node-to-node data transfer, framing, and error detection (e.g., Ethernet, MAC addresses).
- **Layer 1 - Physical:** Transmits raw data bits over physical cables or wireless mediums (e.g., hubs, cables)